answer, but the full story within which it fits is far more subtle and surprising. We'll get there in due course, but we must first make the discussion of this section a touch more precise. And that takes us to the concept of entropy.

## Entropy

Etched into a tombstone in the Zentralfriedhof in Vienna, near the graves of Beethoven, Brahms, Schubert, and Strauss, is a single equation, $S = k \log W$, which expresses the mathematical formulation of a powerful concept known as *entropy*. The tombstone bears the name of Ludwig Boltzmann, one of the most insightful physicists working at the turn of the last century. In 1906, in failing health and suffering from depression, Boltzmann committed suicide while vacationing with his wife and daughter in Italy. Ironically, just a few months later, experiments began to confirm that ideas Boltzmann had spent his life passionately defending were correct.

The notion of entropy was first developed during the industrial revolution by scientists concerned with the operation of furnaces and steam engines, who helped develop the field of thermodynamics. Through many years of research, the underlying ideas were sharply refined, culminating in Boltzmann's approach. His version of entropy, expressed concisely by the equation on his tombstone, uses statistical reasoning to provide a link between the huge number of individual ingredients that make up a physical system and the overall properties the system has.[4]

To get a feel for the ideas, imagine unbinding a copy of *War and Peace*, throwing its 693 double-sided pages high into the air, and then gathering the loose sheets into a neat pile.[5] When you examine the resulting stack, it is enormously more likely that the pages will be out of order than in order. The reason is obvious. There are many ways in which the order of the pages can be jumbled, but only one way for the order to be correct. To be in order, of course, the pages must be arranged precisely as 1, 2; 3, 4; 5, 6; and so on, up to 1,385, 1,386. Any other arrangement is out of order. A simple but essential observation is that, all else being equal, the more ways something can happen, the more likely it